

Experiment No.	3
Experiment Title	Loan Amount Prediction using Regularized Linear Regression 1
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1 Objective

To implement linear regression using Regularization techniques including L1 (Lasso), L2 (Ridge), and L1+L2 (ElasticNet). The experiment also aims to perform hyperparameter tuning to find appropriate alpha penalization values using Cross-Validation, plot the regularized coefficients versus alpha, and plot learning curves for the regularized models.

2 Problem Statement

The problem requires building a robust regression model to predict the continuous `loan_amount` variable based on applicant features such as income, assets, and dependents. The input is tabular demographic and financial data, and the output is a continuous predicted loan amount, scaled appropriately. The objective is to apply regularization to prevent overfitting and improve generalization.

3 Dataset Description

Dataset Name	Loan Approval Dataset (<code>loan_approval_dataset.csv</code>)
Dataset Source	Kaggle
Number of Samples	4,269
Number of Features	12 (Income, Assets, Dependents, Education, etc.)
Number of Classes	N/A (Continuous Target: <code>loan_amount</code>)
Missing Values	Imputed (Categorical with mode, Numerical with median)
Train-Test Split	Train: 2,921 — Validation: 626 — Test: 626

4 Exploratory Data Analysis

Exploratory analysis was performed to understand feature distributions. Visualizations and statistical summaries highlighted that some applicant features (like income and asset values) had significant skew. A total of 96 extreme outlier rows were identified and dropped before scaling the dataset.

5 Data Preprocessing

- **Missing Value Handling:** Numerical columns were imputed with their median. Categorical columns (like Education, Self-Employed, Loan_Status) were imputed using the mode.
- **Encoding:** Categorical string columns were encoded into numeric category codes for matrix compatibility.
- **Feature Scaling:** All numerical continuous columns, including `income_annum`, `cibil_score`, and asset values were standardized (zero mean, unit variance) to ensure regularization penalties were applied uniformly across features.
- **Outlier Removal:** Extreme outliers were removed to prevent distortion in the Ordinary Least Squares (OLS) distance calculations.

6 Machine Learning Algorithm

Four variations of Linear Regression were tested:

- **Standard Linear Regression:** Fits a linear model minimizing the residual sum of squares between the observed targets and predictions. No regularization penalty.
- **Ridge (L2):** Adds an L2 penalty equal to the square of the magnitude of coefficients. Formula: $Cost = RSS + \alpha \sum \beta_i^2$.
- **Lasso (L1):** Adds an L1 penalty equal to the absolute value of the magnitude of coefficients. Can yield sparse models where some coefficients become exactly zero. Formula: $Cost = RSS + \alpha \sum |\beta_i|$.
- **ElasticNet (L1+L2):** A convex combination of Ridge and Lasso penalties, controlled by an `l1_ratio` parameter.

7 Experimental Setup

Python Version	3.10+
Scikit-Learn Version	1.0+
IDE	Jupyter Notebook / VS Code
Operating System	Linux
Hardware	Standard CPU Workspace

8 Hyperparameter Selection

To optimize the regularization models, GridSearchCV was utilized to test various `alpha` penalty weights across cross-validated folds.

Model	Optimal Parameters
Ridge (L2)	<code>alpha = 0.1</code>
Lasso (L1)	<code>alpha = 0.01</code>
ElasticNet	<code>alpha = 0.01, l1_ratio = 0.8</code>

9 Results

1. Test Set Metrics:

Model	MAE	RMSE	R^2 Score
Standard Linear Regression	0.2789	0.3696	0.8736
Ridge Regression	0.2789	0.3696	0.8736
Lasso Regression	0.2801	0.3705	0.8724
ElasticNet Regression	0.2798	0.3701	0.8727

2. 5-Fold Cross Validation (R^2 Mean):

Model	Average CV R^2 Score
Standard Linear Regression	0.8578 (± 0.0127)
Ridge Regression	0.8578 (± 0.0127)
Lasso Regression	0.8580 (± 0.0131)
ElasticNet Regression	0.8578 (± 0.0130)

10 Result Visualization

During the execution, the following graphs were generated to analyze the algorithms:

- **Regularized Coefficients vs Alpha:** Plotted on a logarithmic scale, these plots visually demonstrate how Lasso aggressively pushes coefficient values to exactly zero as alpha increases, performing intrinsic feature selection, whereas Ridge shrinks them asymptotically towards zero. (Saved as `coef_vs_alpha.png`)
- **Learning Curves:** Showcased the training and cross-validation R^2 scores as a function of the number of training examples. The convergence of the curves indicates that the regularized models do not suffer from high variance (overfitting) and stabilize quickly. (Saved as `learning_curves.png`)
- **Actual vs Predicted Scatter Plots:** Displayed the model predictions against actual loan amounts to visualize heteroscedasticity.

11 Discussion

The baseline Linear Regression model performed strongly with an R^2 of 0.8736. Applying Ridge and ElasticNet regularization yielded nearly identical performance, indicating that the baseline model was not severely overfitting. However, applying Lasso regularization revealed an interesting dynamic: despite a negligible drop in test R^2 (to 0.8724), Lasso achieved the highest average Cross-Validation score (0.8580), suggesting that its built-in feature selection slightly improved the model's robustness and generalization capability across unseen data folds.

12 Conclusion

The experiment successfully demonstrated the application of L1 and L2 regularization on a continuous regression task. By plotting the coefficients against varying alpha values, the theoretical difference between Ridge (smooth shrinkage) and Lasso (sparse shrinkage) was practically observed. Furthermore, the learning curves confirmed that the optimal hyperparameter configurations lead to stable and generalizable predictive models for determining loan amounts.

13 References

1. Pedregosa, F., et al. "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, 2011.
2. Tibshirani, R. "Regression Shrinkage and Selection via the Lasso." *Journal of the Royal Statistical Society*, 1996.